

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

3 Economics, Law and Ethics (ah793)

The new Master of (fictional) Porterhouse would like to modernise the college website, using generative AI to create both the html and content. They bring the proposal to the College Council, of which you are a member.

principles of law

- (a) Discuss the challenges presented by intellectual property and generative AI.

[6 marks]

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*Answer:* There are current legal tussles being considered by the courts in relation to intellectual property ownership generated by AI. Cases are being brought before the courts alleging copyright infringement on an unprecedented scale. It may take considerable time for these issues to be decided (*2 marks*). Generative AI is generally trained on data scraped from the web, much of which is copyrighted, and its outputs can resemble those existing works, potentially leading to plagiarism claims (*2 marks*). We are yet to see how these challenges play out. Another copyright challenge is if the generated material is itself covered by copyright, and if so, who owns the copyright (*2 marks*).

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philosophies of ethics

- (b) Use either consequentialist or deontological reasoning to argue in favour of or against the proposal. Additionally, please define the ethical philosophy you will be employing in your argumentation.

[6 marks]

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*Answer:* A good answer will define either consequentialist or deontological ethical philosophies (*2 marks*). Arguments for the proposal include efficiency and innovation (*2 marks*). Arguments against the proposal include concerns that the content and code may create errors, leading to vulnerabilities or inaccuracies that could harm the college's reputation (*2 marks*).

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contemporary ethical issues

- (c) One of the Fellows is concerned about the existential risk of generative AI. Discuss this risk and explain other risks from AI.

[8 marks]

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*Answer:* It is likely their views about the existential risk of AI are linked to the open letter from The Future of Life Institute, which alleges that the risk of AI is comparable to global pandemics and nuclear war (*2 marks*). There has been a moderate amount of pushback against this view that AI presents an existential threat. The concern is that by focusing on remote existential threats, attention is being diverted away from more immediate and pressing concerns (*2 marks*). Large tech companies hold huge amounts of personal information and are using it to manipulate our lives, what to buy, what information to believe, what politics to subscribe to (*2 marks*). Other harms include worker exploitation, massive data theft, the production of synthetic media that reproduces systems of oppression and endangers our information ecosystem, and the concentration of power in the hands of a few people which exacerbates social inequalities (*2 marks*).

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